

Physical Chemistry of Materials

Lecture 23

Proteins



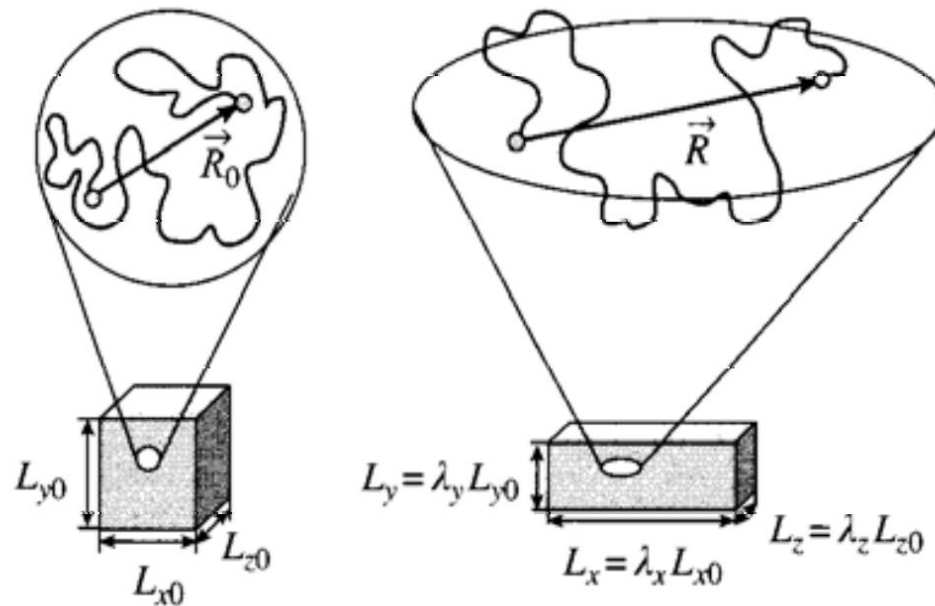
Rubber

- **Rubber:** polymer melt where crosslinks bond the chains together to form a macroscopic network



- “Liquid-like” state (temperature $> T_m$ or T_g)
- Bulk cannot flow due to crosslinks → “soft” solid

Affine Deformation



$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 7.1)

Last Class

Mechanical Properties of Rubber

For density ρ and average molecular mass M_x ,

- Young's modulus

$$E = 3nk_{\text{B}}T = \frac{3\rho RT}{M_x}$$

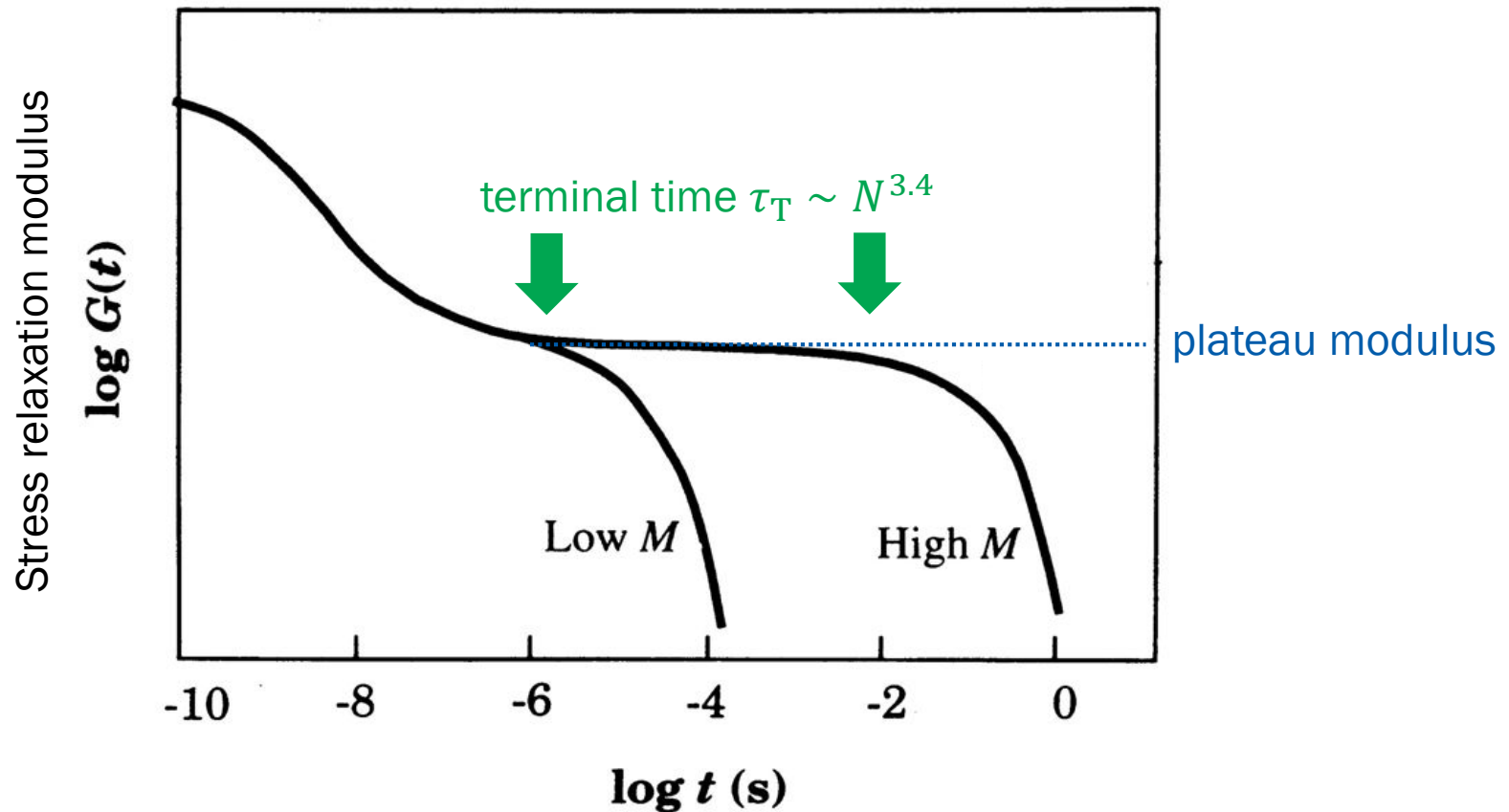
- Shear modulus

$$G = nk_{\text{B}}T = \frac{\rho RT}{M_x}$$

Chemistry does not matter much!



Viscoelasticity of Polymer Melts



(Image: https://link.springer.com/chapter/10.1007/978-3-030-27701-7_20)

Last Class

Viscoelasticity of Polymer Melts

$t < \tau_T$: polymer behaves like an elastic solid

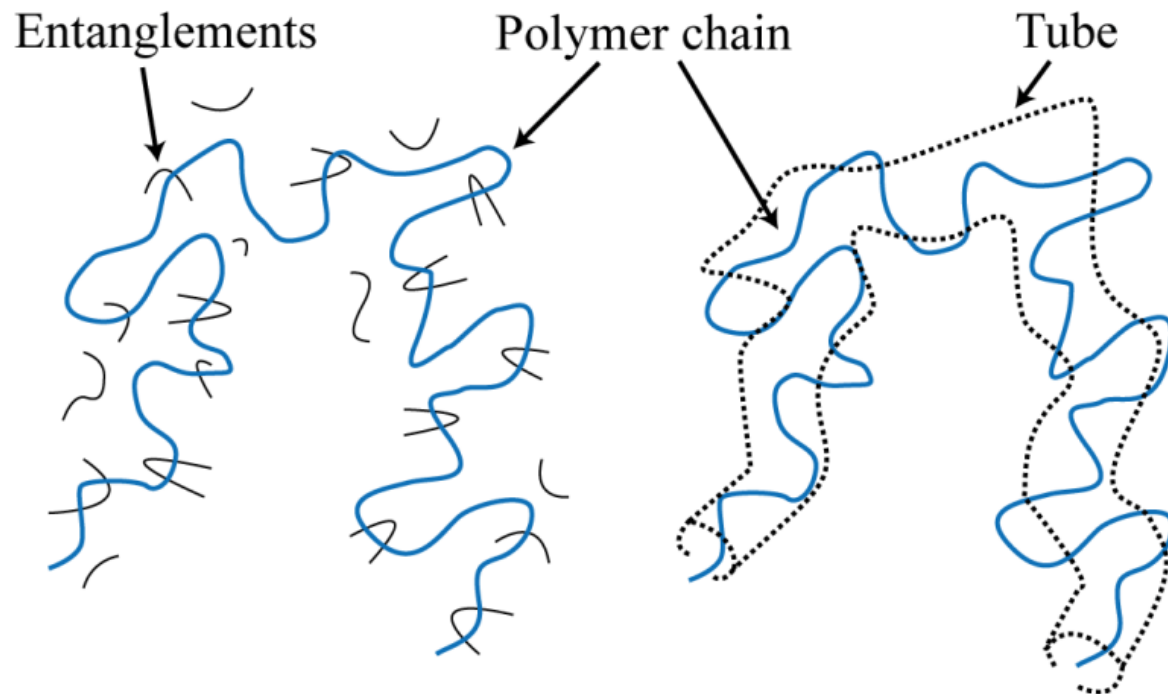
$t > \tau_T$: polymer behaves like a viscous fluid

Why do polymers behave like an elastic solid?

(Note: no crosslinks!)



Viscosity: Tube Model



slithering motion: “reptation”

Prediction vs. Experiment

- Terminal time

$$\tau_T \propto N^3$$

$$(\text{Experiment: } \tau_T \sim N^{3.4})$$

- Self-diffusion coefficient

$$D_{\text{self}} \propto N^{-2}$$

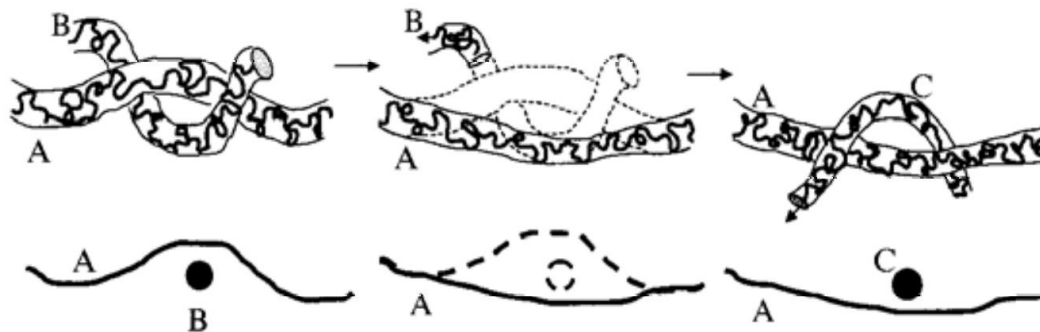
$$(\text{Experiment: } D_{\text{self}} \sim N^{-2.3})$$

Reasons for Discrepancy

- Fluctuation of end-to-end distance

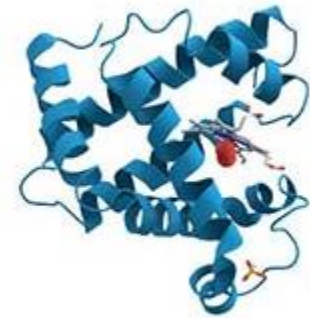
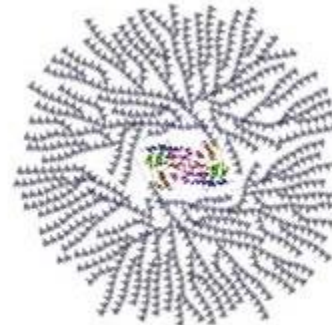
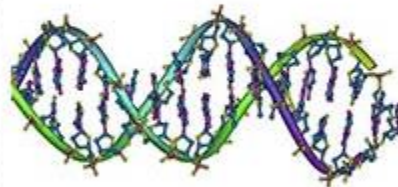


- Reptation of neighboring chains



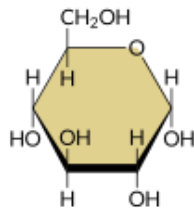
(Image: Rubinstein and Colby, *Polymer Physics* (2003), Figs. 9.1 and 9.25)

Biomolecules

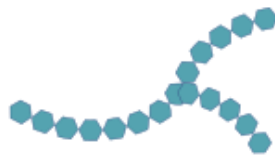


polymers

Heteropolymers



Monosaccharide



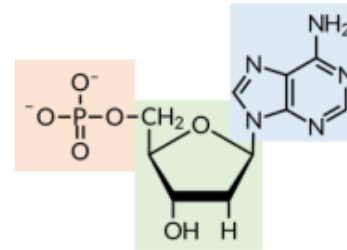
Carbohydrate



Amino acid



Polypeptide

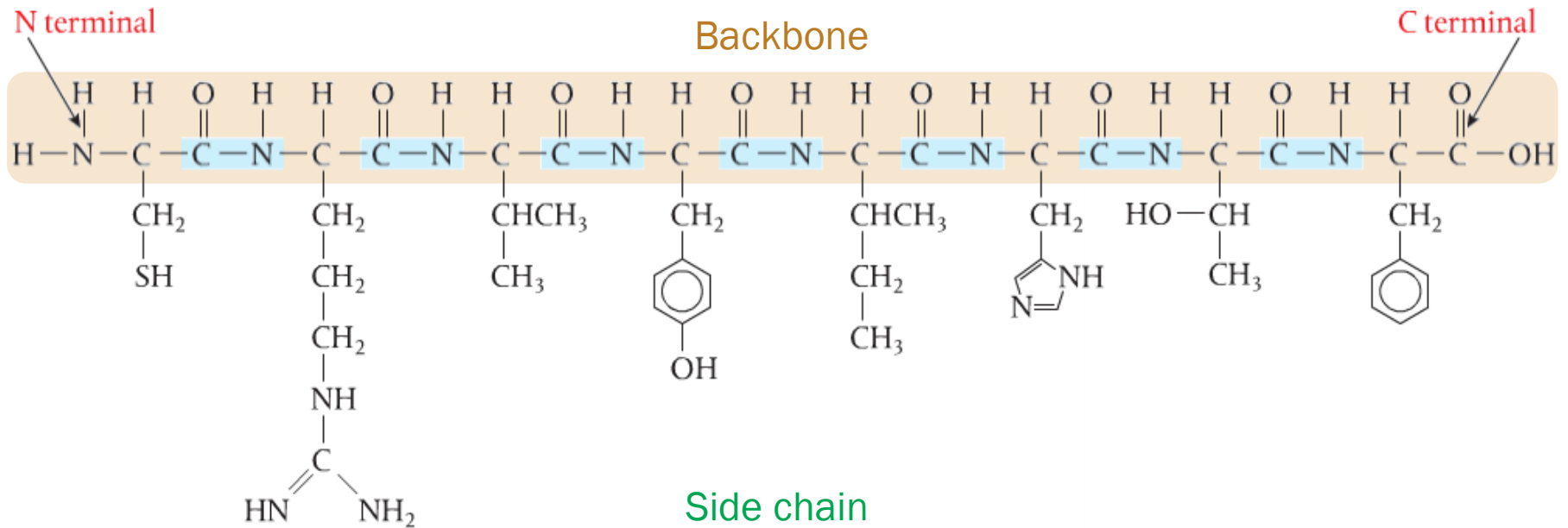


Nucleotide

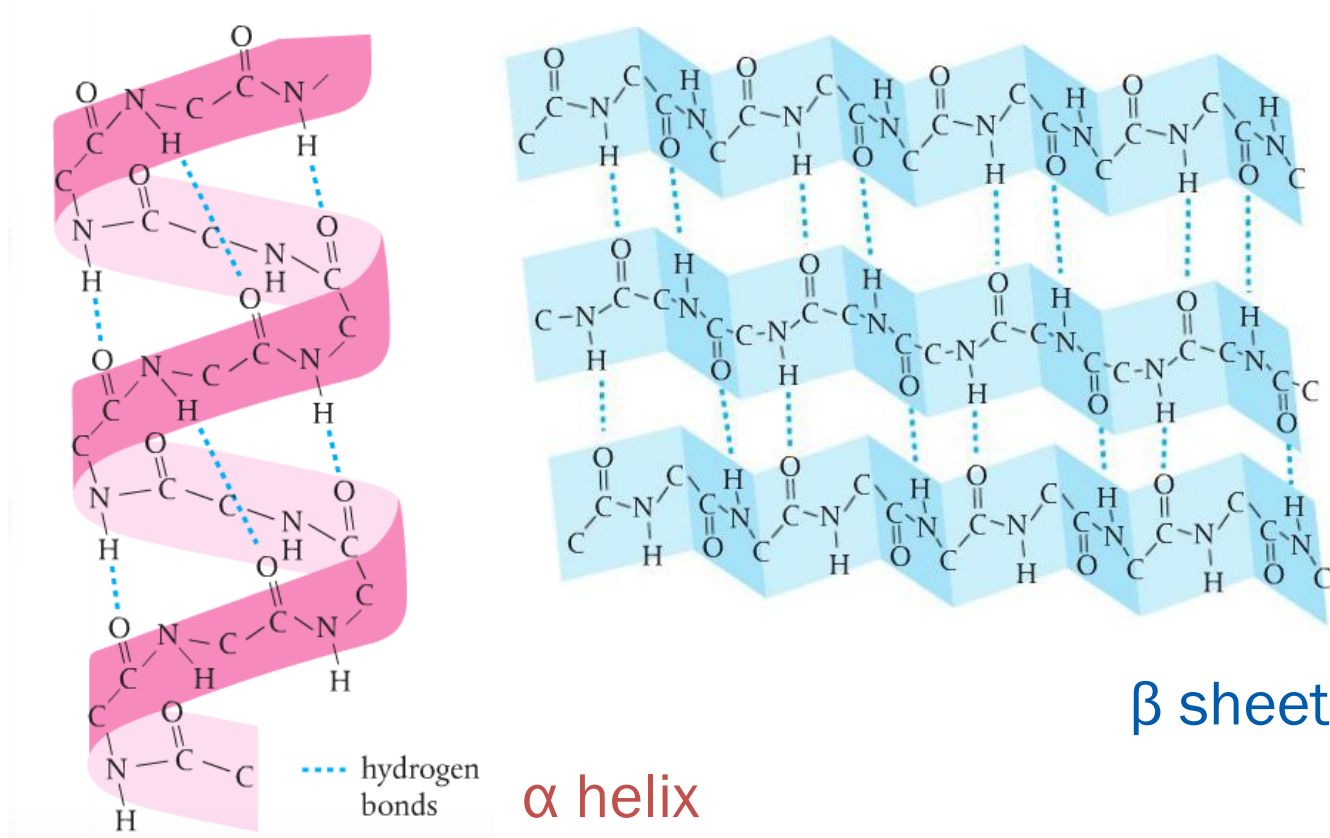


Nucleic acid

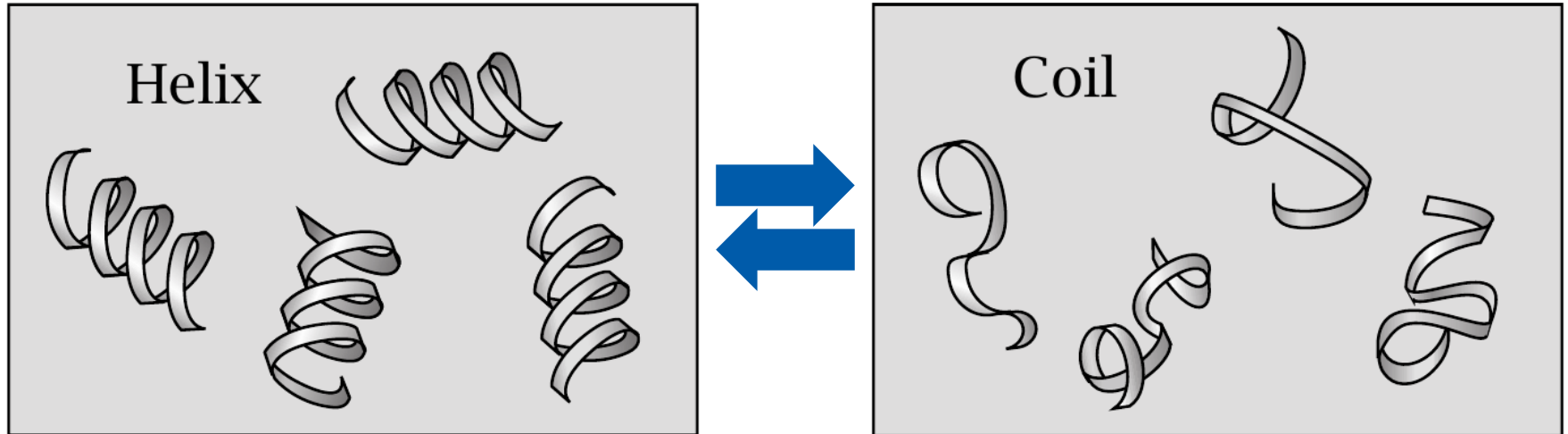
Proteins



Backbone Hydrogen Bonds



Helix-Coil Transition



Modeling HC Transition

- Microstates of a monomer: H or C state

$$\varepsilon = E(\text{H}) - E(\text{C}) < 0$$

- N monomers in each polymer
- Independent monomer model

$$Q = q^N$$

$$q = \sum_i e^{-\beta E_i} = e^{-\beta E(\text{C})} + e^{-\beta E(\text{H})} = e^{-\beta E(\text{C})} (1 + e^{-\beta \varepsilon})$$

Modeling HC Transition

$$Q = q^N$$

$$q = e^{-\beta E(C)} (1 + e^{-\beta \varepsilon})$$

By assuming $E(C) = 0$ and defining $s = e^{-\beta \varepsilon}$,

$$Q = (1 + s)^N$$

The **helicity** is defined as $f_H = \langle n_H \rangle / N$, which is

$$f_H = \frac{e^{-\beta \varepsilon}}{q} = \frac{s}{1 + s}$$

Two-State Model

- Only two polymer states are possible: CCC...C or HHH...H
- Maximum cooperativity!

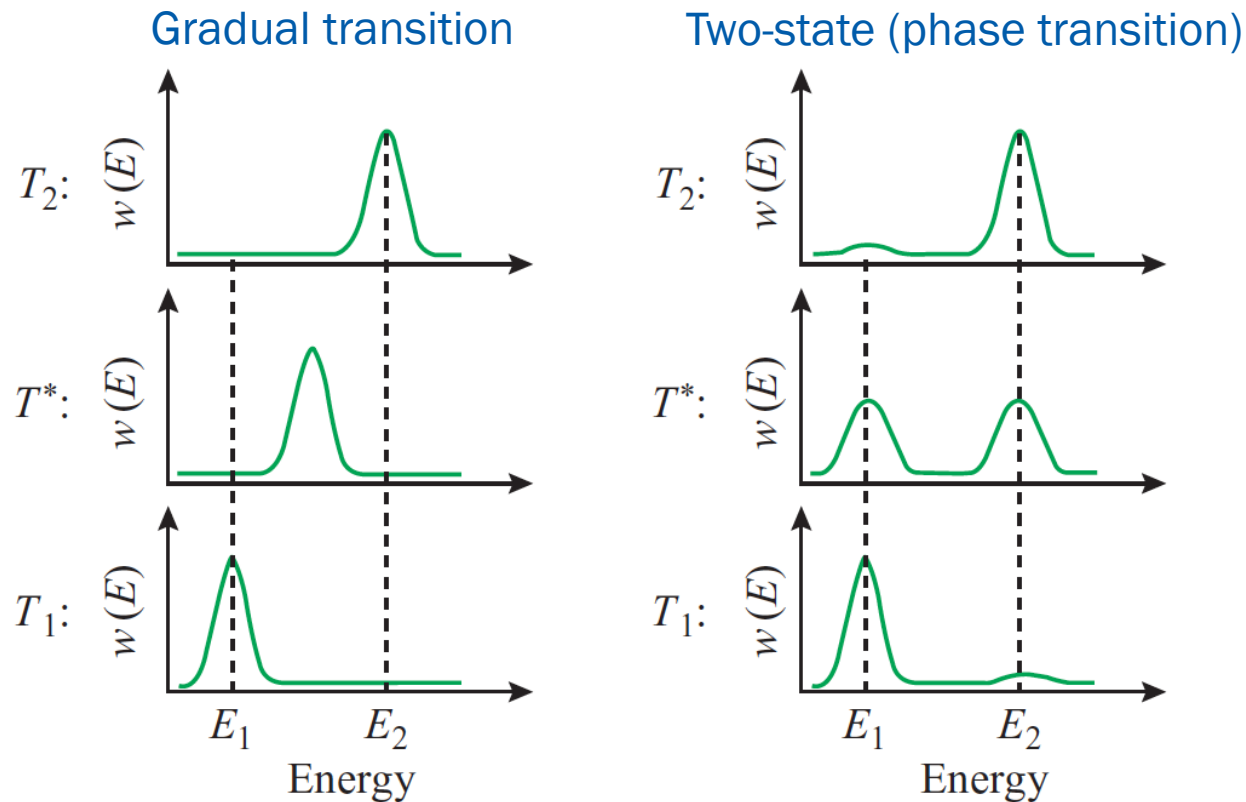
- Partition function

$$Q = q_C^N + q_H^N = e^{-N\beta E(C)} + e^{-N\beta E(H)} = 1 + s^N$$

- Probabilities

$$f_C = \frac{1}{1 + s^N}; \quad f_H = \frac{s^N}{1 + s^N}$$

Different Transitions



(Image: Finkelstein and Ptitsyn, *Protein Physics* (2002), Figs. 8.4 and 8.5)

Zimm-Bragg Model

- Nearest-neighbor cooperativity

$$q(\text{state } j | \text{state } j - 1)$$

- preference for helix: s
- nucleation parameter: σ

state $j - 1$	state j	$q(\text{state } j \text{state } j - 1)$
C	C	$q(C C) = 1$
H	C	$q(C H) = 1$
C	H	$q(H C) = \sigma s$
H	H	$q(H H) = s$

State Vector

- State vector for a monomer

$$\vec{q}_1 = \begin{pmatrix} e^{-\beta E(C)} \\ e^{-\beta E(H)} \end{pmatrix} = \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

- Partition function

$$Q_1 = (1 \quad 1) \vec{q}_1 = (1 \quad 1) \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix} = 1 + \sigma_S$$

Transfer Matrix

$$\vec{q}_1 = \begin{pmatrix} 1 \\ \sigma s \end{pmatrix}$$

- Transfer matrix **T**

$$\mathbf{T} = \begin{pmatrix} q(\text{C}|\text{C}) & q(\text{C}|\text{H}) \\ q(\text{H}|\text{C}) & q(\text{H}|\text{H}) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \sigma s & s \end{pmatrix}$$

- State vector for a dimer

$$\vec{q}_2 = \mathbf{T}\vec{q}_1 = \begin{pmatrix} 1 & 1 \\ \sigma s & s \end{pmatrix} \begin{pmatrix} 1 \\ \sigma s \end{pmatrix} = \begin{pmatrix} 1 + \sigma s \\ \sigma s + \sigma s^2 \end{pmatrix}$$

$$Q_2 = (1 \quad 1)\vec{q}_2 = 1 + 2\sigma s + \sigma s^2$$

Partition Function

$$\vec{q}_1 = \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

$$\mathbf{T} = \begin{pmatrix} q(C|C) & q(C|H) \\ q(H|C) & q(H|H) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \sigma_S & s \end{pmatrix}$$

Generally,

$$\vec{q}_N = \mathbf{T}^{N-1} \vec{q}_1 = \begin{pmatrix} 1 & 1 \\ \sigma_S & s \end{pmatrix}^N \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

$$Q_N = (1 \quad 1) \vec{q}_N = (1 \quad 1) \begin{pmatrix} 1 & 1 \\ \sigma_S & s \end{pmatrix}^N \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

Diagonalization

$$Q_N = (1 \quad 1)\vec{q}_N = (1 \quad 1) \begin{pmatrix} 1 & 1 \\ \sigma_S & s \end{pmatrix}^N \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

Suppose that for $\lambda_1 > \lambda_2$,

$$\begin{pmatrix} 1 & 1 \\ \sigma_S & s \end{pmatrix} = U^{-1} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} U$$

Then

$$Q_N = (1 \quad 1)U^{-1} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^N U \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

Dominant Term

$$Q_N = (1 \quad 1)U^{-1} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^N U \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

$$Q_N = (1 \quad 1)U^{-1} \begin{pmatrix} \lambda_1^N & 0 \\ 0 & \lambda_2^N \end{pmatrix} U \begin{pmatrix} 1 \\ \sigma_S \end{pmatrix}$$

$$Q_N = C_1 \lambda_1^N + C_2 \lambda_2^N$$

For $\lambda_1 > \lambda_2$,

$$Q_N \approx C_1 \lambda_1^N$$

$$\ln Q_N \approx N \ln \lambda_1$$

Largest Eigenvalue

$$\ln Q_N \approx N \ln \lambda_1$$

Now determine λ_1 .

$$\begin{pmatrix} 1 & 1 \\ \sigma s & s \end{pmatrix} = U^{-1} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} U$$

$$\det \begin{vmatrix} 1 - \lambda & 1 \\ \sigma s & s - \lambda \end{vmatrix} = 0$$

$$\lambda^2 - (s + 1)\lambda + s(1 - \sigma) = 0$$

$$\lambda = \frac{(s + 1) \pm \sqrt{(s - 1)^2 + 4s\sigma}}{2}$$

Finally

$$\ln Q_N \approx N \ln \lambda_1$$

$$\lambda = \frac{(s+1) \pm \sqrt{(s-1)^2 + 4s\sigma}}{2}$$

$$\lambda_1 = \frac{(s+1) + \sqrt{(s-1)^2 + 4s\sigma}}{2}$$

$$\ln Q_N \approx N \ln \left[\frac{(s+1) + \sqrt{(s-1)^2 + 4s\sigma}}{2} \right]$$

Helicity

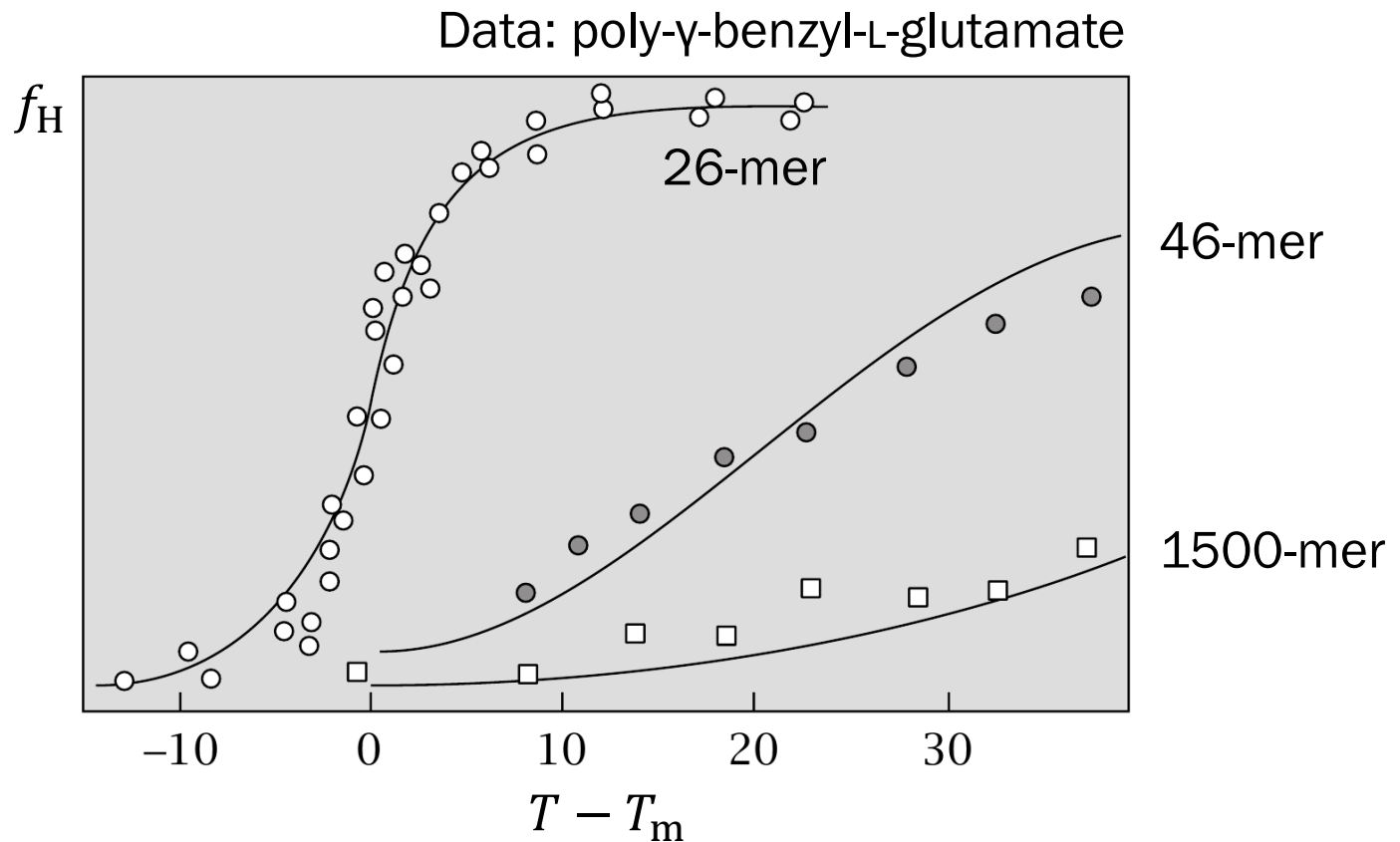
$$f_H = \sum_{\text{seq } i} n_H(\text{seq } i) \frac{e^{-\beta E(\text{seq } i)}}{Q_N}$$

$$f_H = \sum_{\text{seq } i} n_H(\text{seq } i) \frac{e^{-\beta \varepsilon n_H(\text{seq } i) - \dots}}{Q_N}$$

$$f_H = \frac{1}{Q_N} \sum_{\text{seq } i} -\frac{1}{\beta} \frac{\partial [e^{-\beta \varepsilon n_H(\text{seq } i) - \dots}]}{\partial \varepsilon} = -\frac{1}{\beta Q_N} \frac{\partial Q_N}{\partial \varepsilon}$$

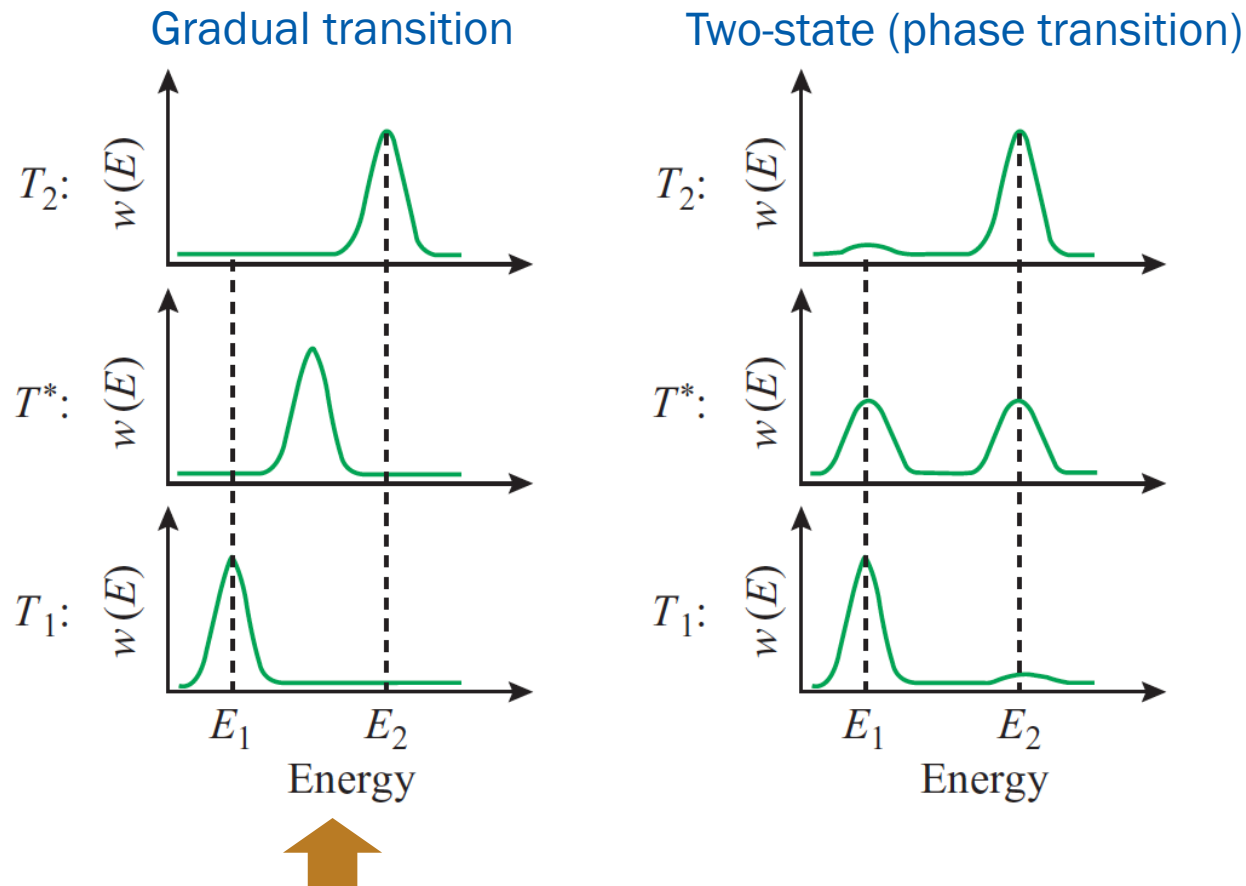
$$f_H = -\frac{1}{\beta} \frac{\partial \ln Q_N}{\partial \varepsilon} = -\frac{1}{\beta} \frac{\partial \ln Q_N}{\partial s} \frac{ds}{d\varepsilon}$$

Experimental Data



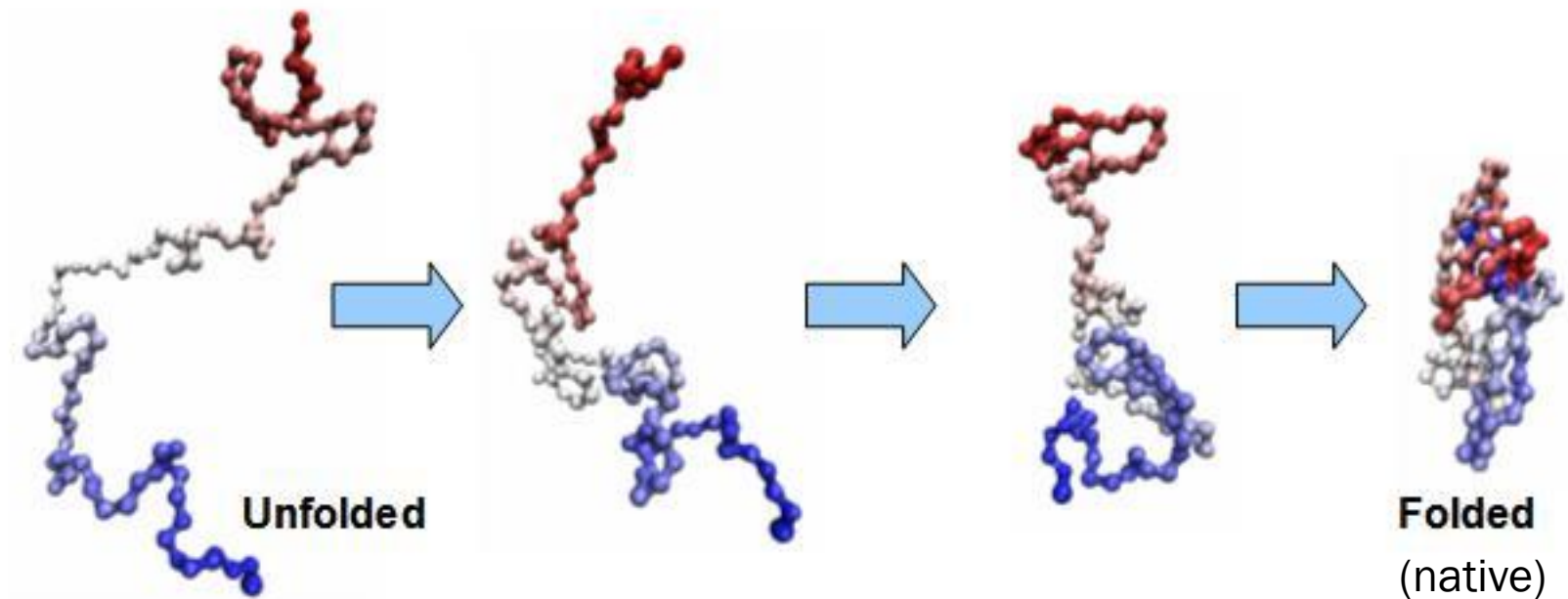
(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Fig. 26.11)

Phase Transition in ZB Model?

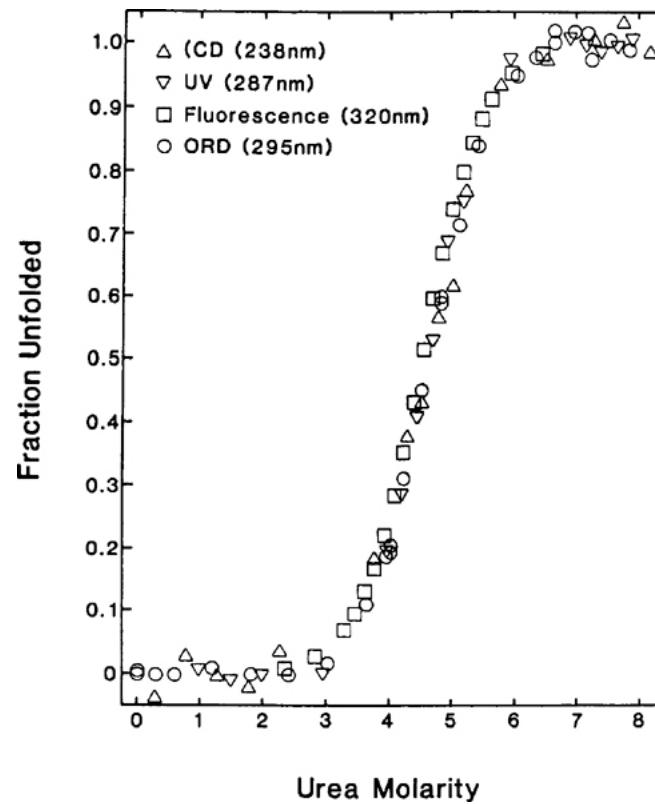


(Image: Finkelstein and Ptitsyn, *Protein Physics* (2002), Figs. 8.4 and 8.5)

Protein Folding



Unfolding Curve



van't Hoff Criterion

Assuming two-state folding,

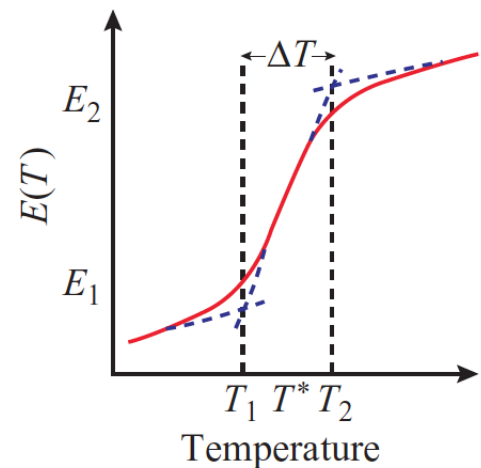
$$P_{\text{fold}} = \frac{e^{-\beta G_{\text{folded}}}}{e^{-\beta G_{\text{folded}}} + e^{-\beta G_{\text{unfolded}}}} = \frac{1}{1 + e^{\beta \Delta G_{\text{fold}}}}$$

Using the Gibbs-Helmholtz equation,

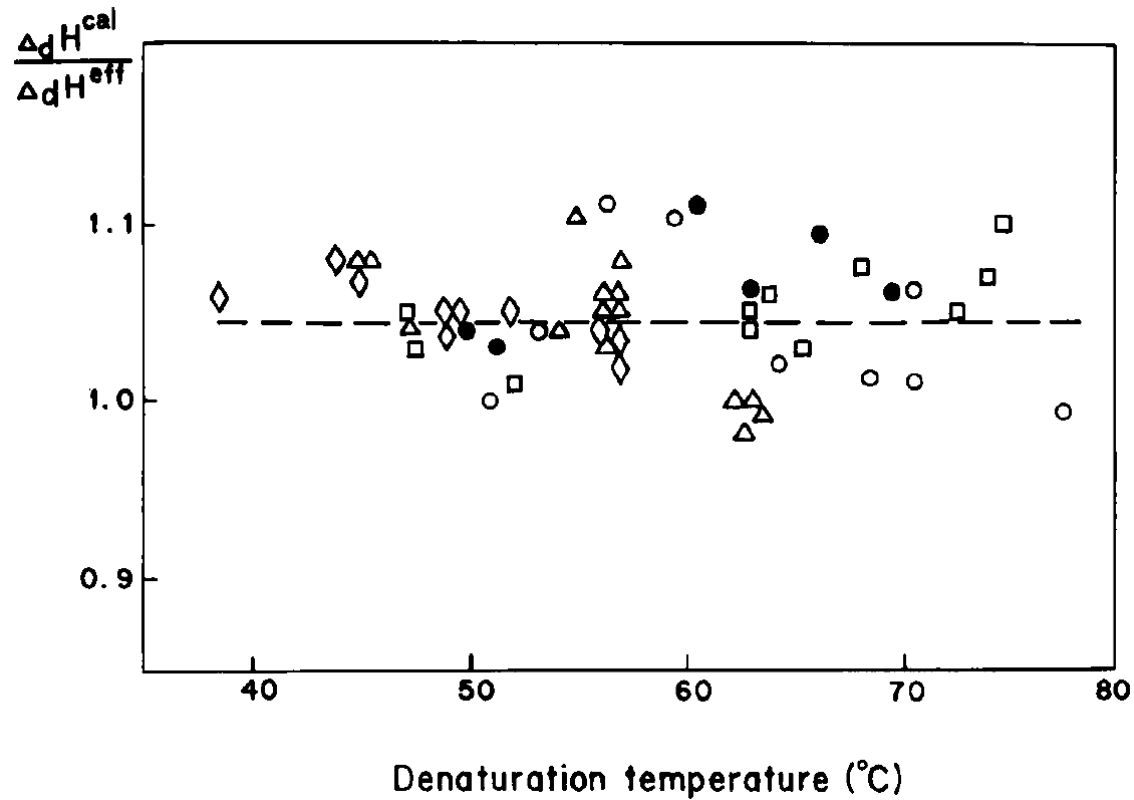
$$\frac{dP_{\text{fold}}}{dT} = P_{\text{fold}}(1 - P_{\text{fold}}) \frac{\Delta H_{\text{fold}}}{k_B T^2}$$

At the transition point ($P_{\text{fold}} = 1/2$),

$$\left. \frac{dP_{\text{fold}}}{dT} \right|_{T=T^*} = \frac{\Delta H_{\text{fold}}}{4k_B (T^*)^2}$$

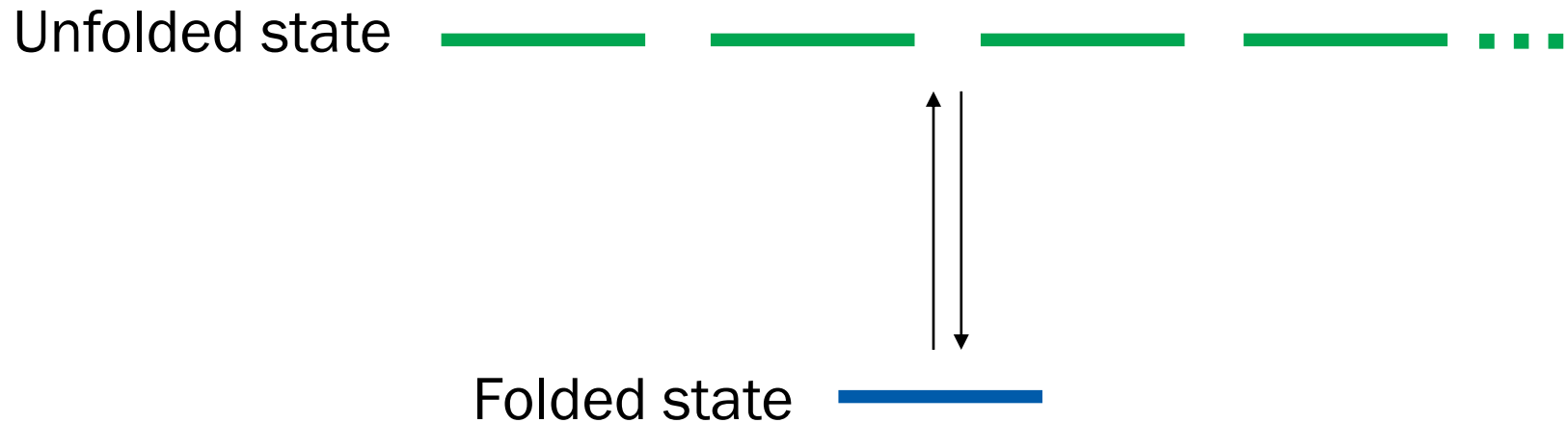


Experimentally ...



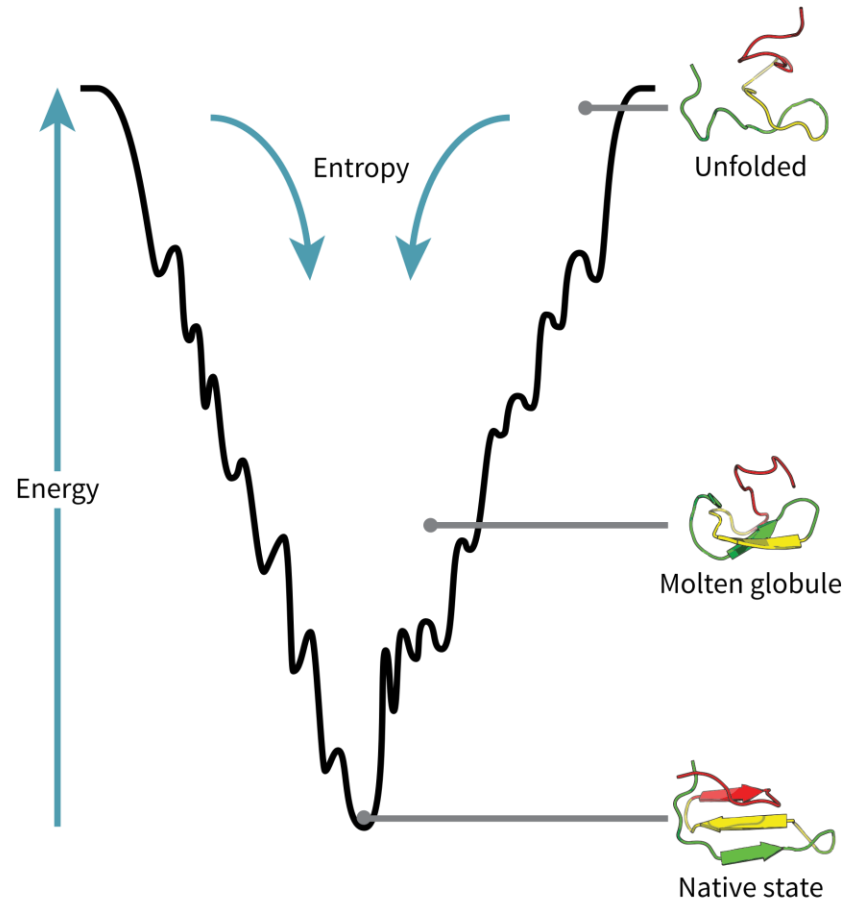
(Image: Privalov, *Adv Prot Chem* (1979))

Two-State Model



What is the underlying physics?

“Funnel” Picture



(Image: https://en.wikipedia.org/wiki/Protein_folding#/media/File:Folding_funnel_schematic.svg)